Finding and Verifying Inverse Functions
Answer Key
Algebra 2

Quick Answers

1. $\frac{x+8}{5}$
2. $4x - 12$
3. (a) $= x$, —
4. $\frac{3x+1}{x-2}$
5. $(x+2)^2 - 7$
6. 4
7. $\frac{3-5x}{2}$
8. $\frac{4}{x-1} - 6$
9. x
10. (a) $= \frac{7-x}{3}$, (b) $= x$
11. (a) $= x$, —
12. (a) $= \frac{x+3}{1-2x}$, —

What is verified

13 of 16 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 3, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSF-BF.A.1c – *problems 3, 9, 10, 11*
HSF-BF.B.4 – *problems 1, 2, 4, 5, 6, 7, 8, 10, 11, 12*
Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

6. *If they got 1.75: undid the operations in the wrong order: divided by 4 first, then subtracted 3*

1. Find the inverse rule for $f(x) = 5x - 8$.
Solution: Write $x = f(y)$ and solve for y.

$$\begin{aligned} x &= 5y - 8 \\ x + 8 &= 5y \\ y &= \frac{x+8}{5} \end{aligned}$$

Every step is reversible, so the rule found is the inverse. $\boxed{f^{-1}(x) = \frac{x+8}{5}}$

---

2. Find the inverse rule for $g(x) = \dfrac{x}{4} + 3$.

Solution: Write $x = g(y)$ and undo the two operations in reverse order.

$$\begin{aligned} x &= \frac{y}{4} + 3 \\ x - 3 &= \frac{y}{4} \\ y &= 4(x-3) = 4x - 12 \end{aligned}$$

Check: $g(4x-12) = \frac{4x-12}{4} + 3 = x - 3 + 3 = x$. $\boxed{g^{-1}(x) = 4x - 12}$

---

3. Let $p(x) = 2x + 9$ and $q(x) = \dfrac{x-9}{2}$. (a) Simplify $p(q(x))$. (b) Explain what the result says about p and q.

Solution (a): Substitute $q(x)$ into p and simplify.

$$\begin{aligned} p\big(q(x)\big) &= 2\left(\frac{x-9}{2}\right) + 9 \\ &= (x-9) + 9 = x \end{aligned}$$

The composition collapses to the identity. $\boxed{x}$

Solution (b) — instructor-judged. A correct answer says that p and q undo each other: feeding a number through q and then through p returns the original number, so q is the inverse of p (and p the inverse of q). Look for the word *inverse* (or "they undo each other") together with a reference to the part (a) result being x. Saying only "they are related" or "they are opposites" is not enough. A student who adds that $q(p(x)) = x$ must also hold has given the complete two-sided argument — worth extra credit, not required.

---

4. Find the inverse rule for $r(x) = \dfrac{2x+1}{x-3}$.

Solution: Write $x = r(y)$, clear the fraction, then collect every y on one side.

$$\begin{aligned} x &= \frac{2y+1}{y-3} \\ x(y-3) &= 2y + 1 \\ xy - 3x &= 2y + 1 \\ xy - 2y &= 3x + 1 \\ y(x-2) &= 3x + 1 \end{aligned}$$

Dividing by $x - 2$ finishes it. $\boxed{r^{-1}(x) = \dfrac{3x+1}{x-2}}$

---

5. Find the inverse rule for $s(x) = \sqrt{x+7} - 2$, inputs $x \geq -7$.

Solution: Isolate the radical first, then square.

$$\begin{aligned} x &= \sqrt{y+7} - 2 \\ x + 2 &= \sqrt{y+7} \\ (x+2)^2 &= y + 7 \\ y &= (x+2)^2 - 7 \end{aligned}$$

Squaring is safe because $\sqrt{\ }$ is never negative, so $x + 2 \geq 0$ on the range of s.

$$\boxed{s^{-1}(x) = (x+2)^2 - 7}$$

---

6. For $m(x) = 4x + 3$, find $m^{-1}(19)$.
Solution: Build the inverse rule, then evaluate it at 19.

$$\begin{aligned} x = 4y + 3 \quad &\Longrightarrow \quad y = \frac{x-3}{4} \\ m^{-1}(19) = \frac{19-3}{4} &= \frac{16}{4} \end{aligned}$$

Equivalently, ask what input makes $4x + 3 = 19$. $\boxed{m^{-1}(19) = 4}$

---

7. Find the inverse rule for $t(x) = \dfrac{3-2x}{5}$.
Solution: Multiply by 5 before touching the numerator.

$$\begin{aligned} x &= \frac{3-2y}{5} \\ 5x &= 3 - 2y \\ 2y &= 3 - 5x \end{aligned}$$

The sign of the x term flips because y was subtracted. $\boxed{t^{-1}(x) = \dfrac{3-5x}{2}}$

---

8. Find the inverse rule for $u(x) = \dfrac{4}{x+6} + 1$.
Solution: Peel the outside operations off in reverse order, then reciprocate.

$$\begin{aligned} x &= \frac{4}{y+6} + 1 \\ x - 1 &= \frac{4}{y+6} \\ y + 6 &= \frac{4}{x-1} \\ y &= \frac{4}{x-1} - 6 \end{aligned}$$

The input $x = 1$ is excluded, matching the horizontal asymptote $y = 1$ of u.

$$\boxed{u^{-1}(x) = \frac{4}{x-1} - 6}$$

9. Let $w(x) = (x-4)^3+2$. Test the claim $w^{-1}(x) = \sqrt[3]{x-2}+4$ by simplifying $w\big(w^{-1}(x)\big)$.
Solution: Substitute the claimed rule into w and simplify.

$$\begin{aligned} w\big(w^{-1}(x)\big) &= \Big(\big(\sqrt[3]{x-2}+4\big)-4\Big)^3+2 \\ &= \big(\sqrt[3]{x-2}\big)^3+2 \\ &= (x-2)+2 = x \end{aligned}$$

The composition returns the input, so the classmate is right. $\boxed{x}$

10. Let $n(x) = 7-3x$. (a) Find $n^{-1}(x)$. (b) Simplify $n\big(n^{-1}(x)\big)$.
Solution (a):

$$\begin{aligned} x &= 7-3y \\ 3y &= 7-x \\ y &= \frac{7-x}{3} \end{aligned}$$

Watch the subtraction: $3y$ moves across, not $-3y$. $\boxed{n^{-1}(x) = \dfrac{7-x}{3}}$

Solution (b):

$$\begin{aligned} n\big(n^{-1}(x)\big) &= 7-3\left(\frac{7-x}{3}\right) \\ &= 7-(7-x) = x \end{aligned}$$

The identity confirms part (a). $\boxed{x}$

11. Let $v(x) = x^2-6x+5$ with $x \geq 3$. (a) Confirm $v^{-1}(x) = 3+\sqrt{x+4}$. (b) Explain why the restriction is needed.
Solution (a): Completing the square first makes the check short: $v(x) = (x-3)^2-4$.

$$\begin{aligned} v\big(v^{-1}(x)\big) &= \Big(\big(3+\sqrt{x+4}\big)-3\Big)^2-4 \\ &= \big(\sqrt{x+4}\big)^2-4 \\ &= (x+4)-4 = x \end{aligned}$$

The claimed rule really does undo v. $\boxed{x}$

Solution (b) — instructor-judged. Look for the one-to-one idea. Without a restriction v is a parabola with vertex $(3,-4)$, so it takes every output twice — one input on each side of $x=3$ (for example $v(1) = v(5) = 0$). A rule that sends 0 back to a single input cannot exist, so v has no inverse on all of its natural domain. Restricting to $x \geq 3$ keeps only the increasing half, which is one-to-one; that half is also why the inverse takes the $+\sqrt{\ }$ branch and not $3-\sqrt{x+4}$. Full credit: the one-to-one/fails-horizontal-line idea plus a reference to the vertex or to two inputs sharing an output.

---

12. Let $f(x) = \dfrac{x-3}{2x+1}$. (a) Find $f^{-1}(x)$. (b) Name the excluded input and explain.

Solution (a): Clear the fraction and collect the y terms.

$$\begin{aligned} x(2y+1) &= y-3 \\ 2xy + x &= y-3 \\ 2xy - y &= -x-3 \\ y(2x-1) &= -(x+3) \\ y &= \frac{x+3}{1-2x} \end{aligned}$$

Multiplying numerator and denominator by -1 gives the tidier form above.

$$\boxed{f^{-1}(x) = \frac{x+3}{1-2x}}$$

Solution (b) — instructor-judged. The answer is $x = \frac{1}{2}$. Accept any response that names that value and says the denominator $1-2x$ is zero there, so the expression is undefined. The deeper reason, worth extra credit: $\frac{1}{2}$ is the horizontal asymptote of f, so it is the one output f never actually produces — an input the inverse can never be asked about. Do not accept "$x = -\frac{1}{2}$" (that is the exclusion for f itself, not for f^{-1}).
